

Bayesian Prior Justification Report

Clinical Trial

Biostatistics

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1 Executive Summary

This report documents the Bayesian prior specification for study **Clinical Trial**, prepared by **Biostatistics** (Sponsor) and dated 08 May 2026.

2 Trial Information

Field	Value
Trial / Protocol	Clinical Trial
Sponsor	Sponsor
Statistician	Biostatistics
Report date	08 May 2026

3 Prior Specification

3.1 Elicitation Methodology

3.2 Fitted Parameters

3.3 Prior Summary Statistics

3.4 Prior Density

4 Prior-Data Conflict Assessment

Warning: No prior-data conflict diagnostics were computed for this report.

4.1 Prior - Likelihood - Posterior Overlay

5 Sensitivity Analysis

Warning: No sensitivity analysis was conducted for this report.

5.1 Influence of Prior Hyperparameters

5.2 Tornado Plot

5.3 Hyperparameter Influence Heatmap

6 Robust and Sensitivity Priors

Warning: No robust, sceptical, or power priors were computed for this report.

7 Regulatory Compliance Checklist

Table 1: FDA/EMA regulatory compliance checklist

Requirement	Status
Prior elicitation method documented	Missing
Distribution family and parameters specified	Missing
Expert / source of prior identified	Missing
Prior density plot provided	Missing
Prior-data conflict assessed	Not run
Conflict diagnostic statistics reported	Not run
Sensitivity analysis performed	Not run
Sensitivity visualisations provided	Not run
Alternative priors considered	Not run
Robust / sceptical prior computed	Not run
Regulatory report generated	Complete

Note: This checklist maps to requirements in the FDA Draft Guidance on Bayesian Statistical Methods (2026) and the EMA Reflection Paper on the Use of Bayesian Statistics in the Evaluation of Medicinal Products.

8 Appendix: Session Information

Item	Value
R version	4.6.0
bayprior version	0.1.0
quarto R package	1.5.1
Quarto CLI version	1.9.37
Platform	x86_64-pc-linux-gnu
Date	08 May 2026

*Report generated by the **bayprior** R package using `prior_report()`. Methods follow O'Hagan et al. (2006), Box (1980), Schmidli et al. (2014), Ibrahim & Chen (2000), and the FDA Draft Guidance on Bayesian Methods in Clinical Trials (2026).*